

## ENEE452 HW09 : Software System Schematic Representation

Figure 1 is a portion of a larger diagram representing a MQTT-based system for remote data acquisition. When the system schematic is fully realized it will show a network of processing nodes and their interactions which the system architect has created to document the system design.

Elements of the diagram:

L475 (the B-L475-IOT1A micro) and ISM43362 (Inventek radio) nodes have buffer functions (1 and 1) which pass 'signals' (configuration info, data, event notifications) over a link between them.

L475 also has 'settings' (x,y,z) representing information which the system architect has about other parts of the design.

As the system design evolves the system architect's job is to identify which features of the software systems and multipurpose modules are needed for the design and how they are to be configured and connected. This information will be recorded in footnotes on the system schematic. When the schematic is complete it will be the roadmap for constructing the system.

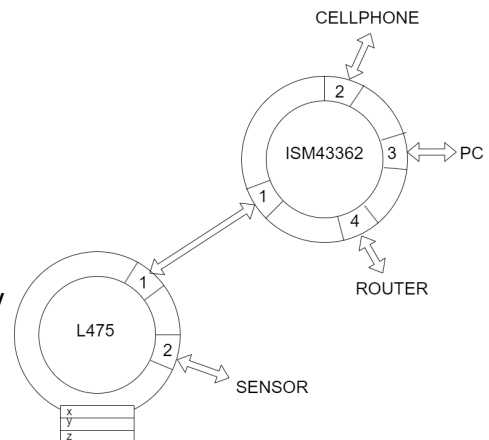


Figure 1

**Problem 1)** Identify configuration values and the buffer 1 signals which will be required for L475 to establish a connection to the internet through one of ISM43362's internet access options (cellphone, PC, or router). By 'signals' we mean identification of things which will be transmitted through the buffer functions, not the buffer functions themselves. The 'values' to be reported are abstract identifications like "server website IP" not the literal URL of some target website.

A guess on the way forward in this problem is to find example code which does the job, inspect it to determine which functions in (a usually very large) library of functions are being called, and then record the arguments which those functions are processing. Knowing what the function needs to process in our specific situation is our primary goal. If we know this specific information then (hopefully) we can avoid getting tangled up in all of the other use-cases which those types of libraries and functions are designed to handle.